

SEO

I. Introduction – What Is SEO

Search Engine Optimization (SEO) is often considered the more technical part of Web marketing. This is true because SEO does help in the promotion of sites and at the same time it requires some technical knowledge – at least familiarity with basic HTML. SEO is sometimes also called SEO copyrighting because most of the techniques that are used to promote sites in search engines deal with text. Generally, SEO can be defined as the activity of optimizing Web pages or whole sites in order to make them more search engine-friendly, thus getting higher positions in search results.

One of the basic truths in SEO is that even if you do all the things that are necessary to do, this does not automatically guarantee you top ratings but if you neglect basic rules, this certainly will not go unnoticed. Also, if you set realistic goals – i.e to get into the top 30 results in Google for a particular keyword, rather than be the number one for 10 keywords in 5 search engines, you will feel happier and more satisfied with your results.

Although SEO helps to increase the traffic to one's site, SEO is not advertising. Of course, you can be included in paid search results for given keywords but basically the idea behind the SEO techniques is to get top placement because your site is relevant to a particular search term, not because you pay.

SEO can be a 30-minute job or a permanent activity. Sometimes it is enough to do some generic SEO in order to get high in search engines – for instance, if you are a leader for rare keywords, then you do not have a lot to do in order to get decent placement. But in most cases, if you really want to be at the top, you need to pay special attention to SEO and devote significant amounts of time and effort to it. Even if you plan to do some basic SEO, it is essential that you understand how search engines work and which items are most important in SEO.

1. How Search Engines Work

The first basic truth you need to learn about SEO is that search engines are not humans. While this might be obvious for everybody, the differences between how humans and search engines view web pages aren't. Unlike humans, search engines are text-driven. Although technology advances rapidly, search engines are far from intelligent creatures that can feel the beauty of a cool design or enjoy the sounds and movement in movies. Instead, search engines crawl the Web, looking at particular site items (mainly text) to get an idea what a site is about. This brief explanation is not the most precise because as we will see next, search engines perform several activities in order to deliver search results – *crawling*, *indexing*, *processing*, *calculating relevancy*, and *retrieving*.

First, search engines **crawl** the Web to see what is there. This task is performed by a piece of software, called a *crawler* or a *spider* (or Googlebot, as is the case with Google). Spiders follow links from one page to another and index everything they find on their way. Having in mind the number of pages on the Web (over 20 billion), it is impossible for a spider to visit a site daily just

to see if a new page has appeared or if an existing page has been modified. Sometimes crawlers will not visit your site for a month or two, so during this time your SEO efforts will not be rewarded. But there is nothing you can do about it, so just keep quiet.

What you can do is to check what a crawler sees from your site. As already mentioned, crawlers are not humans and they do not see images, Flash movies, JavaScript, frames, password-protected pages and directories, so if you have tons of these on your site, you'd better run the **Spider Simulator** below to see if these goodies are viewable by the spider. If they are not viewable, they will not be spidered, not indexed, not processed, etc. - in a word they will be non-existent for search engines.

After a page is crawled, the next step is to **index** its content. The indexed page is stored in a giant database, from where it can later be retrieved. Essentially, the process of indexing is identifying the words and expressions that best describe the page and assigning the page to particular keywords. For a human it will not be possible to process such amounts of information but generally search engines deal just fine with this task. Sometimes they might not get the meaning of a page right but if you help them by optimizing it, it will be easier for them to classify your pages correctly and for you – to get higher rankings.

When a search request comes, the search engine **processes** it – i.e. it compares the search string in the search request with the indexed pages in the database. Since it is likely that more than one pages (practically it is millions of pages) contains the search string, the search engine starts **calculating the relevancy** of each of the pages in its index to the search string.

There are various algorithms to calculate relevancy. Each of these algorithms has different relative weights for common factors like keyword density, links, or metatags. That is why different search engines give different search results pages for the same search string. What is more, it is a known fact that all major search engines, like Yahoo!, Google, MSN, etc. periodically change their algorithms and if you want to keep at the top, you also need to adapt your pages to the latest changes. This is one reason (the other is your competitors) to devote permanent efforts to SEO, if you'd like to be at the top.

The last step in search engines' activity is **retrieving** the results. Basically, it is nothing more than simply displaying them in the browser – i.e. the endless pages of search results that are sorted from the most relevant to the least relevant sites.

2. Differences Between the Major Search Engines

Although the basic principle of operation of all search engines is the same, the minor differences between them lead to major changes in results relevancy. For different search engines different factors are important. There were times, when SEO experts joked that the algorithms of Yahoo! are intentionally made just the opposite of those of Google. While this might have a grain of truth, it is a matter of fact that the major search engines like different stuff and if you plan to conquer more than one of them, you need to optimize carefully.

There are many examples of the differences between search engines. For instance, for Yahoo! and MSN, on-page keyword factors are of primary importance, while for Google links are very, very important. Also, for Google sites are like wine – the older, the better, while Yahoo! generally has no expressed preference towards sites and domains with tradition (i.e. older ones). Thus you might need more time till your site gets mature to be admitted to the top in Google, than in Yahoo!.

II. Keywords – the Most Important Item in SEO

Keywords are the most important SEO item for every search engine – actually they are what search strings are matched against. So you see that it is very important that you optimize your site for the right keywords. This seems easy at first but when you get into more detail, it might be a bit confusing to correctly determine the keywords. But with a little research and thinking the problem of selecting the right keywords to optimize for can be solved.

1. Choosing the Right Keywords to Optimize For

It seems that the time when you could easily top the results for a one-word search string is centuries ago. Now, when the Web is so densely populated with sites, it is next to impossible to achieve constant top ratings for a one-word search string. Achieving constant top ratings for two-word or three-word search strings is a more realistic goal. If you examine closely the dynamics of search results for popular one-word keywords, you might notice that it is so easy one week to be in the first ten results and the next one – to have fallen out of the first 30 results because the competition for popular one-word keywords is so fierce and other sites have replaced you.

Of course, you can include one-word strings in your keywords list but if they are not backed up by more expressions, do not dream of high ratings. For instance, if you have a site about dogs, “*dog*” is a mandatory keyword but if you do not optimize for more words, like “*dog owners*”, “*dog breeds*”, “*dog food*”, or even “*canine*”, success is unlikely, especially for such a popular keyword. The examples given here are by no means the ultimate truth about how to optimize a dog site but they are good enough to show that you need to think broad when choosing the keywords.

Generally, when you start optimization, the first thing you need to consider is the keywords that describe the content of your site best and that are most likely to be used by users to find you. Ideally, you know your users well and can guess correctly what search strings they are likely to use to search for you. One issue to consider is synonyms. Very often users will use a different word for the same thing. For instance, in the example with the dog site, “*canine*” is a synonym and it is for sure that there will be users who will use it, so it does not hurt to include it now and then on your pages. But do not rush to optimize for every synonym you can think of – search engines themselves have algorithms that include synonyms in the keyword match, especially for languages like English.

Instead, think of more keywords that are likely to be used to describe your site. Thinking thematically is especially good because search engines tend to rate a page higher if it belongs to a site the theme of which fits into the keyword string. In this aspect it is important that your site is concentrated around a particular theme – i.e. dogs. It might be difficult to think of all the relevant keywords on your own but that is why tools are for. You can also try Google's [Keyword Tool](#) to get more suggestions about which keywords are hot and which are not.

When choosing the keywords to optimize for, you need to consider not only their relevancy to your site and the expected monthly number of searches for these particular keywords. Very often narrow searches are more valuable because the users that come to your site are those that are really interested in your product. If we go on with the dog example, you might discover that the “adopt a dog” keyphrase brings you more visitors because you have a special section on your site where you give advice on what to look for when adopting a dog. This page is not of interest of current dog owners but to potential dog owners only, who might be not so many in number but are your target audience and the overall effect of attracting this niche can be better than attracting everybody who is interested in dogs in general. So, when you look at the numbers of search hits per month, consider the unique hits that fit into the theme of your site.

2. Keyword Density

After you have chosen the keywords that describe your site and are supposedly of interest to your users, the next step is to make your site keyword-rich and to have good *keyword density* for your target keywords. Keyword density is a common measure of how relevant a page is. Generally, the idea is that the higher the keyword density, the more relevant to the search string a page is. The recommended density is 3-7% for the major 2 or 3 keywords and 1-2% for minor keywords.

Although there are no strict rules, try optimizing for a reasonable number of keywords – 5 or 10 is OK. If you attempt to optimize for a list of 300, you will soon see that it is just not possible to have a good keyword density for more than a few keywords, without making the text sound artificial and stuffed with keywords. And what is worse, there are severe penalties (including ban from the search engine) for *keyword stuffing* because this is considered an unethical practice that tries to manipulate search results.

3. Keywords in Special Places

Keywords are very important not only as quantity but as quality as well – i.e. if you have more keywords in the page title, the headings, the first paragraphs – this counts more than if you have many keywords at the bottom of the page. The reason is that the URL (and especially the domain name), file names and directory names, the page title, the headings for the separate sections are more important than ordinary text on the page and therefore, all equal, if you have the same keyword density as your competitors but you have keywords in the URL, this will boost your ranking incredibly, especially with Yahoo!.

a. Keywords in URLs and File Names

The domain name and the whole URL of a site tell a lot about it. The presumption is that if your site is about dogs, you will have “dog”, “dogs”, or “puppy” as part of your domain name. For instance, if your site is mainly about adopting dogs, it is much better to name your dog site “dog-adopt.net” than “animal-care.org”, for example, because in the first case you have two major keywords in the URL, while in the second one you have no more than one potential minor keyword.

When hunting for keyword rich domain names, don't get greedy. While from a SEO point of view it is better to have 5 keywords in the URL, just imagine how long and difficult to memorize the URL will be. So you need to strike a balance between the keywords in the URL and site usability, which says that more than 3 words in the URL is a way too much. Probably you will not be able to come on your own with tons of good suggestions. Additionally, even if you manage to think of a couple of good domain names, they might be already taken.

File names and directory names are also important. Often search engines will give preference to pages that have a keyword in the file name. For instance <http://mydomain.com/dog-adopt.html> is not as good as <http://dog-adopt.net/dog-adopt.html> but is certainly better than <http://mydomain.com/animal-care.html>. The advantage of keywords in file names over keywords in URLs is that they are easier to change, if you decide to move to another niche, for example.

b. Keywords in Page Titles

The page title is another special place because the contents of the <title> tag usually gets displayed in most search engines, (including Google). While it is not mandatory per the HTML specification to write something in the <title> tag (i.e. you can leave it empty and the title bar of the browser will read “Untitled Document” or similar), for SEO purposes you may not want to leave the <title> tag empty; instead, you'd better write the the page title in it.

Unlike URLs, with page titles you can get wordy. If we go on with the dog example, the <title> tag of the home page for the <http://dog-adopt.net> can include something like this: <title>Adopt a Dog – Save a Life and Bring Joy to Your Home</title>, <title>Everything You Need to Know About Adopting a Dog</title> or even longer.

c. Keywords in Headings

Normally headings separate paragraphs into related subtopics and from a literary point of view, it may be pointless to have a heading after every other paragraph but from SEO point of view it is extremely good to have as many headings on a page as possible, especially if they have the keywords in them.

There are no technical length limits for the contents of the <h1>, <h2>, <h3>, ... <hN> tags but common sense says that too long headings are bad for page readability. So, like with URLs, you need to be wise with the length of headings. Another issue you need to consider is how the heading will be displayed. If it is Heading 1 (<h1>), generally this means larger font size and in

this case it is recommendable to have less than 7-8 words in the heading, otherwise it might spread on 2 or 3 lines, which is not good and if you can avoid it – do it.

III. Links – Another Important SEO Item

1. Why Links Are Important

Probably the word that associates best with Web is “links”. That is what hypertext is all about – you link to pages you like and get linked by pages that like your site. Actually, the Web is woven out of interconnected pages and spiders follow the links, when indexing the Web. If not many sites link to you, then it might take ages for search engines to find your site and even if they find you, it is unlikely that you will have high rankings because the quality and quantity of links is part of the algorithms of search engines for calculating relevancy.

2. Inbound and Outbound Links

Put in layman's terms, there are two types of links that are important for SEO – inbound and outbound links. *Outbound links* are links that start from your site and lead to another one, while *inbound links*, or *backlinks*, come from an external site to yours, e.g. if a.com links to mydomain.com, the link from a.com is an inbound link for mydomain.com.

Backlinks are very important because they are supposed to be a measure of the popularity of your site among the Web audience. It is necessary to say that not all backlinks are equal. There are good and bad backlinks. Good backlinks are from reputable places - preferably from sites with a similar theme. These links do boost search engine ranking. Bad backlinks come from suspicious places – like link farms – and are something to be avoided. Well, if you are backlinked without your knowledge and consent, maybe you should drop the Webmaster a line, asking him or her to remove the backlink.

If you are not heavily backlinked, don't worry - buying links is an established practice and if you are serious about getting to the top, you may need to consider it. But before doing this, you should consider some free alternatives. For instance, some of the good places where you can get quality backlinks are Web directories like <http://dmoz.org> or <http://dir.yahoo.com>.

First, look for suitable sites to backlink to you using the **Backlinks Builder** below. After you identify potential backlinks, it's time to contact the Web master of the site and to start negotiating terms. Sometimes you can agree to a barter deal – i.e. a link exchange – they will put on their site N links to your site and you will put on your site N links to their site - but have in mind that this is a bad, risky deal and you should always try to avoid it

Internal links (i.e. links from one page to another page on the same site) are also important but not as much as backlinks. In this connection it is necessary to say, that using images for links might be prettier but it is a SEO killer. Instead of having buttons for links, use simple text links.

Since search engines spider the text on a page, they can't see all the designer miracles, like gradient buttons or flash animations, so when possible, either avoid using them, or provide a meaningful textual description in the <alt> tag, as described next.

3. Anchor text

Anchor text is the most important item in a backlink. While it does matter where a link comes from (i.e. a reputable place or a link farm), what matters more is the actual text the link starts from. Put simply, *anchor text* is the word(s) that you click on to open the hyperlink – e.g. if we have [the best search engine](#), then “the best search engine” is the anchor text for the hyperlink to google.com. You see that you might have a backlink from a valuable site but if the anchor text is something like “an example of a complete failure”, you will hardly be happy with it.

When you check your backlinks, always check what their anchor text is and if there is a keyword in it. It is a great SEO boost to have a lot of backlinks from quality sites and the anchor text to include our keywords. Check the anchor text of inbound backlinks is with the **Backlink Anchor Text Analyzer** tool below. Besides the anchor text itself, the text around it is also important.

4. Link Practices That Are To Be Avoided

Similar to keyword stuffing, purchasing links in bulk is a practice to be avoided. It gets suspicious if you bartered 1000 links with another site in a day or two. What is more, search engines keep track of link farms (sites that sell links in bulk) and since bought links are a way to manipulate search results, this practice gets punished by search engines. So avoid dealing with link farms because it can cause more harm than do good. Also, outbound links from your site to known Web spammers or “bad guys” are also to be avoided.

As mentioned, link exchange is not a clean deal. Even if it boosts your ranking, it can have many other negative aspects in the long run. First, you do not know if the other party will keep their promise – i.e. they might remove some of the links to you. Second, they might change the context the link appears into. Third, it is really suspicious if you seem to be “married” to another site and 50% or more of your inbound and outbound links are from/to this direction.

When links are concerned, one aspect to have in mind is the ratio between inbound and outbound links. Generally speaking, if your outbound links are ten times your inbound links, this is bad but it also varies on a case by case basis. If you have a site that links to news sources or has RSS feeds, then having many outbound links is the inevitable price of fresh content.

IV. Metatags

A couple of years ago <meta> tags were the primary tool for search engine optimization and there was a direct correlation between what you wrote there and your position in search results. However, algorithms got better and today the importance of metadata is decreasing day by day, especially with Google. But still some search engines show metadata (under the clickable link in search results), so users can read what you have written and if they think it is relevant, they

might go to your site. Also, some of the specialized search engines still use the metatags when ranking your site.

The meta Description tag is one more way for you to write a description of your site, thus pointing search engines to what themes and topics your Web site is relevant to. It does not hurt to include at least a brief description, so don't skip it. For instance, for the dog adoption site, the meta Description tag could be something like this: `<Meta Name="Description" Content="Adopting a dog saves a life and brings joy to your house. All you need to know when you consider adopting a dog.">`

A potential use of the meta Keywords tags is to include a list of keywords that you think are relevant to your pages. The major search engines will not take this into account but still it is a chance for you to emphasize your target keywords. You may consider including alternative spellings (or even common misspellings of your keywords) in the meta Keywords tag. For instance, if I were to write the meta keywords tag for the dog adoption site, I would do it like that: `<Meta name="Keywords" Content="adopt, adoption, dog, dogs, puppy, canine, save a life, homeless animals">`. It is a small boost to search engine top ranking but why miss the chance?

The meta Robots tag deserves more attention. In this tag you specify the pages that you do NOT want crawled and indexed. It happens that on your site you have contents that you need to keep there but you don't want it indexed. Listing this pages in the meta Robots tag is one way to exclude them (the other way is by using a robots.txt file and generally this is the better way to do it) from being indexed.

V. Content Is King

If you are new to SEO, it might be a surprise for you that text is one of the driving forces to higher rankings. But it is a fact. Search engines (and your readers) love fresh content and providing them with regularly updated, relevant content is a recipe for success. Generally, when a site is frequently updated, this increases the probability that the spider will revisit the site sooner. You can't take for sure that if you update your site daily, the spider will visit it even once a week but if you do not update your contents regularly, this will certainly drop you to from the top of search results.

For company sites that are not focused on writing but on manufacturing constantly adding text can be a problem because generally company sites are not reading rooms or online magazines that update their content daily, weekly or monthly but even for company sites there are reasonable solutions. No matter what your business is, one is for sure – it is always relevant to include a news section on your site – it can be company news or RSS feeds but this will keep the ball rolling.

1. Topical Themes or How to Frequently Add Content to Your Site

If you are doing the SEO for an online magazine, you can consider yourself lucky – fresh content is coming all the time and you just need to occasionally arrange a heading or two or a couple of

paragraphs to make the site SEO-friendly. But even if you are doing a SEO for an ordinary company site, it is not all that bad - there are ways to constantly get fresh content that fits into the topic of the site.

One of the intricacies of optimizing a company site is that it has to be serious. Also, if your content smells like advertising and has no practical value for your visitors, this content is not that valuable. For instance, if you are a trade company, you can have promotional texts about your products. But have in mind that these texts must be informational, not just sales hype. And if you have a lot of products to sell, or frequently get new products, or make periodical promotions of particular products and product groups – you can post all this to your site and you will have fresh, topical content.

Also, depending on what your business is about, you can include different kinds of self-updating information like lists of hot new products, featured products, discounted items, even online calculators or order trackers. Unlike promotional pages, this might neither bring you many new visitors, nor improve your ratings but is more than nothing.

One more potential traffic trigger for company sites are news sections. Here you can include news about past and coming events, post reports about various activities, announce new undertakings, etc. Some companies even go further – their CEO keeps a blog, where he or she writes in a more informal style about what is going in the company, in the industry as a whole, or in the world in general. These blogs do attract readers, especially if the information is true, rather than the official story.

An alternative way to get fresh free content are RSS feeds. RSS feeds are gaining more and more popularity and with a little bit of searching, you can get free syndicated content for almost any topic you can think of.

2. Bold and Italic Text

When you have lots of text, the next question is how to make the important items stand out from the crowd – for both humans and search engines. While search engines (and their spiders – the programs that crawl the Web and index pages) cannot read text the way humans do, they do have ways of getting the meaning of a piece of text. Headings are one possibility, bold and italic are another way to emphasize a word or a couple of words that are important. Search engines read the and <i> text and get the idea that what is in bold and/or italic is more important than the rest of the text. But do not use bold and italic too much – this will spoil the effect, rather than make the whole page a search engine favorite.

3. Duplicate Content

When you get new content, there is one important issue – is this content original? Because if it is not, i.e. it is stolen from another site, this will get you into trouble. But even if it is not illegal, i.e. you obtained it for free from an article feed, have in mind that you might not be only one on the Web, who has this particular stuff. If you have the rights to do it, you can change the text a little, so it is not an exact copy of another page and cannot be labeled “duplicate content” by search

engines. If you don't manage to escape the duplicate content filter that search engines have imposed recently in their attempts to filter stolen, scrapped, or simply copied contents, your pages could be removed from search results!

Duplicate content became an issue when tricky webmasters started making multiple copies of the same page (under a different name) in order to fool search engines that they have more content than they actually do. As a result of this malpractice, search engines responded with a duplicate content filter that removes suspicious pages. Unfortunately, this filter sometimes removes quite legitimate pages, like product descriptions given from a manufacturer to all its resellers, which must be kept exactly the same.

You see, duplicate content can be a serious problem. But it is not an obstacle that cannot be overcome. First, you need to periodically check the Web for pages that are similar to yours. You can use <http://copyscape.com>. If you identify pages that are similar to yours (and it is not you who have illegitimately copied them), you could notify the webmaster of the respective site(s) to remove them. Also, you could change a little the text on your site, hoping that this way you will avoid the duplicate content penalty. Even with product descriptions, you can add commentary or opinion on the same page and this could be a way out.

VI. Visual Extras and SEO

As already mentioned, search engines have no means to index directly extras like images, sounds, flash movies, javascript. Instead, they rely on you to provide meaningful textual description and based on it they can index these files. In a sense, the situation is similar to that with text 10 or so years ago – you provide a description in the metatag and search engines uses this description to index and process your page. If technology advances further, one day it might be possible for search engines to index images, movies, etc. but for the time being this is just a dream.

1. Images

Images are an essential part of any Web page and from a designer point of view they are not an extra but a most mandatory item for every site. However, here designers and search engines are on two poles because for search engines every piece of information that is buried in an image is lost. When working with designers, sometimes it takes a while to explain to them that having textual links (with proper anchor text) instead of shining images is not a whim and that clear text navigation is really mandatory. Yes, it can be hard to find the right balance between artistic performance and SEO-friendliness but since even the finest site is lost in cyberspace if it cannot be found by search engines, a compromise to its visual appearance cannot be avoided.

With all that said, the idea is not to skip images at all. Sure, nowadays this is impossible because the result would be a most ugly site. Rather the idea is that images should be used for illustration and decoration, not for navigation or even worse – for displaying text (in a fancy font, for example). And the most important – in the <alt> attribute of the tag, always provide a meaningful textual description of the image. The HTML specification does not require this but

search engines do. Also, it does not hurt to give meaningful names to the image files themselves rather than name them image1.jpg, image2.jpg, imageN.jpg. For instance, in the next example the image file has an informative name and the alt provides enough additional information: ``. Well, don't go to extremes like writing 20-word `<alt>` tags for 1 pixel images because this also looks suspicious and starts to smell like keyword-stuffing.

2. Animation and Movies

The situation with animation and movies is similar to that with images – they are valuable from a designer's point of view but are not loved by search engines. For instance, it is still pretty common to have an impressive Flash introduction on the home page. You just cannot imagine what a disadvantage with search engines this is – it is a number one rankings killer! And it gets even worse, if you use Flash to tell a story that can be written in plain text, hence crawled and indexed by search engines. One workaround is to provide search engines with a HTML version of the Flash movie but in this case make sure that you have excluded the original Flash movie from indexing (this is done in the robots.txt file but the explanation of this file is not a beginners topic and that is why it is excluded from this tutorial), otherwise you can be penalized for duplicate content.

There are rumors that Google is building a new search technology that will allow to search inside animation and movies and that the .swf format will contain new metadata that can be used by search engines, but until then, you'd better either refrain from using (too much) Flash, or at least provide a textual description of the movie (you can use an `<alt>` tag to describe the movie).

3. Frames

It is a good news that frames are slowly but surely disappearing from the Web. 5 or 10 years ago they were an absolute hit with designers but never with search engines. Search engines have difficulties indexing framed pages because the URL of the page is the same, no matter which of the separate frames is open. For search engines this was a shock because actually there were 3 or 4 pages and only one URL, while for search engines 1 URL is 1 page. Of course, search engines can follow the links to the pages in the frameset and index them but this is a hurdle for them.

If you still insist on using frames, make sure that you provide a meaningful description of the site in the `<noframes>` tag. The following example is not for beginners but even if you do not understand everything in it, just remember that the `<noframes>` tag is the place to provide an alternative version (or at least a short description) of your site for search engines and users whose browsers do not support frames. If you decide to use the `<noframes>` tag, maybe you'd like to read more about it before you start using it.

Example: `<noframes> <p> This site is best viewed in a browser that supports frames. </p><p> Welcome to our site for prospective dog adopters! Adopting a homeless dog is a most noble deed that will help save the life of the poor creature. </p></noframes>`

4. JavaScript

This is another hot potato. It is known by everybody that pure HTML is powerless to make complex sites with a lot of functionality (anyway, HTML was not intended to be a programming language for building Web applications, so nobody expects that you can use HTML to handle writing to a database or even for storing session information) as required by today's Web users and that is why other programming languages (like JavaScript, or PHP) come to enhance HTML. For now search engines just ignore JavaScript they encounter on a page. As a result of this, first if you have links that are inside the JavaScript code, chances are that they will not be spidered. Second, if JavaScript is in the HTML file itself (rather than in an external .js file that is invoked when necessary) this clutters the html file itself and spiders might just skip it and move to the next site. Just for your information, there is a `<noscript>` tag that allows to provide alternative to running the script in the browser but because most of its applications are pretty complicated, it is hardly suitable to explain it here.

VII. Static Versus Dynamic URLs

Based on the previous section, you might have gotten the impression that the algorithms of search engines try to humiliate every designer effort to make a site gorgeous. Well, it has been explained why search engines do not like image, movies, applets and other extras. Now, you might think that search engines are far too cheeky to dislike dynamic URLs either. Honestly, users are also not in love with URLs like <http://domain.com/product.php?cid=1&pid=5> because such URLs do not tell much about the contents of the page.

There are a couple of good reasons why static URLs score better than dynamic URLs. First, dynamic URLs are not always there – i.e. the page is generated on request after the user performs some kind of action (fills a form and submits it or performs a search using the site's search engine). In a sense, such pages are nonexistent for search engines, because they index the Web by crawling it, not by filling in forms.

Second, even if a dynamic page has already been generated by a previous user request and is stored on the server, search engines might just skip it if it has too many question marks and other special symbols in it. Once upon a time search engines did not index dynamic pages at all, while today they do index them but generally slower than they index static pages.

The idea is not to revert to static HTML only. Database-driven sites are great but it will be much better if you serve your pages to the search engines and users in a format they can easily handle. One of the solutions of the dynamic URLs problem is called *URL rewriting*. There are special tools (different for different platforms and servers) that rewrite URLs in a friendlier format, so they appear in the browser like normal HTML pages. Try the **URL Rewriting Tool** below, it will convert the cryptic text from the previous example into something more readable, like <http://mydomain.com/product-categoryid-1-productid-5>.

VIII. Promoting Your Site to Increase Traffic

The main purpose of SEO is to make your site visible to search engines, thus leading to higher rankings in search results pages, which in turn brings more traffic to your site. And having more visitors (and above all buyers) is ultimately the goal in sites promotion. For truth's sake, SEO is only one alternative to promote your site and increase traffic – there are many other online and offline ways to do accomplish the goal of getting high traffic and reaching your target audience. We are not going to explore them in this tutorial but just keep in mind that search engines are not the only way to get visitors to your site, although they seem to be a preferable choice and a relatively easy way to do it.

1. Submitting Your Site to Search Directories, forums and special sites

After you have finished optimizing your new site, time comes to submit it to search engines. Generally, with search engines you don't have to do anything special in order to get your site included in their indices – they will come and find you. Well, it cannot be said exactly when they will visit your site for the first time and at what intervals they will visit it later but there is hardly anything that you can do to invite them. Sure, you can go to their Submit a Site pages in submit the URL of your new site but by doing this do not expect that they will hop to you right away. What is more, even if you submit your URL, most search engines reserve the right to judge whether to crawl your site or not. Anyway, here are the URLs for submitting pages in the three major search engines: [Google](#), [MSN](#), and [Yahoo](#).

In addition to search engines, you may also want to have your site included in search directories as well. Although *search directories* also list sites that are relevant to a given topic, they are different from search engines in several aspects. First, search directories are usually maintained by humans and the sites in them are reviewed for relevancy after they have been submitted. Second, search directories do not use crawlers to get URLs, so *you* need to go to them and submit your site but once you do this, you can stay there forever and no more efforts on your side are necessary. Some of the most popular search directories are DMOZ and Yahoo! (the directory, not the search engine itself) and here are the URLs of their submissions pages: [DMOZ](#) and [Yahoo!](#).

Sometimes posting a link to your site in the right forums or special sites can do miracles in terms of traffic. You need to find the forums and sites that are leaders in the fields of interest to you but generally even a simple search in Google or the other major search engines will retrieve their names. For instance, if you are a hardware freak, type “hardware forums” in the search box and in a second you will have a list of sites that are favorites to other hardware freaks. Then you need to check the sites one by one because some of them might not allow posting links to commercial sites. Posting into forums is more time-consuming than submitting to search engines but it could also be pretty rewarding.

2. Specialized Search Engines

Google, Yahoo!, and MSN are not the only search engines on Earth, nor even the only general-purpose ones. There are many other general-purpose and specialized search engines and some of them can be really helpful for reaching your target audience. You just can't imagine for how many niches specialized search engines exist – from law, to radiostations, to educational one!

Some of them are actually huge sites that gather Webwide resources on a particular topic but almost all of them have sections for submitting links to external sites of interest. So, after you find the specialized search engines in your niche, go to their site and submit your URL – this could prove more trafficworthy than striving to get to the top of Google.

3. Paid Ads and Submissions

We have already mentioned some other alternatives to search engines – forums, specialized sites and search engines, search directories – but if you need to make sure that your site will be noticed, you can always resort to paid ads and submissions. Yes, paid listings are a fast and guaranteed way to appear in search results and most of the major search engines accept payment to put your URL in the Paid Links section for keywords of interest to you but you also must have in mind that users generally do not trust paid links as much as they do with the normal ones – in a sense it looks like you are bribing the search engine to place you where you can't get on your own, so think twice about the pros and cons of paying to get listed.